

Role Support in Knowledge-Graph Error Ranking: Predictor Regimes and Evaluation Policies

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A knowledge graph may never have seen an entity in a particular relation role. Does that make a new prediction unreliable? This paper shows why the answer depends on the predictor and the decision being evaluated, while identifying settings where learning from role counts remains useful.

The question

Knowledge-graph completion predicts a missing entity in a triple. An error ranker then orders predictions for acceptance or review. The study asks whether relation-specific training support improves that ranking beyond confidence, frequency, and ensemble disagreement. Its outcome is recovery of recorded held-out answers: an unrecorded true fact can count as an error, so the experiment does not measure factual falsehood.

The approach

Head-role and tail-role counts measure how often the relevant entity appeared in that relation role in training. Missing-role bits indicate a zero count. Features are evaluated at the predicted tail, without using hidden answers. The experiments cover 12 from-scratch DistMult/ComplEx predictors, six continued-training stages, and six fixed three-member ensembles on FB15k-237 and WN18RR. Logistic and nonlinear error heads compare nested feature sets on fixed predictions; candidate and acceptance policies are tested separately.

The main result

On FB15k-237, a continued ComplEx regime raises mean recorded-target accuracy from 10.05% to 44.35%. Yet the raw positive missing-role penalty reverses its error-ranking direction: mean AUROC falls from 0.6893 to 0.3639. This comparison changes multiple training choices and predicted answers; it does not isolate accuracy as the cause. Learned counts still help. Adding continuous role counts beyond confidence, frequency, and explicit disagreement improves AURC (error across answer rates) and Brier (error-probability scoring) in all 12 fixed-ensemble/error-head comparisons. Gains do not imply lower selective risk at every answer rate.

Why the evaluation policy matters

A missing-role bit is a deterministic function of its count, so it adds no conditional information; it can still change how a restricted scorer fits. At an 85% batch answer rate, count-based logistic models reach the hindsight error floor in all original runs but none of the continued-training runs. Candidate restrictions also have mixed effects: training-supported tails improve DistMult accuracy on FB15k-237 but reduce it on WN18RR, where 27.67% of queries lose every recorded target. Development-fixed thresholds produce unequal test answer rates, changing the tradeoff being compared.

What this contributes

The contribution is an inspectable evaluation of when established support features help and when their interpretation fails. It does not establish a new uncertainty decomposition or calibrated factual truth. The two static graphs were repeatedly inspected; optimizer seeds and warm starts are not independent graph replications. New graphs, temporal transfer, more complete answer labels, and reproduction of historical results remain open.

KEY TAKEAWAY

Treat role support as a predictor-dependent feature. Judge it against strong comparators under the exact candidate, error, and acceptance policy that matters.